ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** a simplified representation of a system that keeps what matters for the question	Models leave out unimportant details.
2	**(a)** a single kind of particle with fixed composition	Pure substances have fixed composition.
3	**(b)** Antoine Lavoisier	Lavoisier.
4	**(b)** electron	Cathode ray experiments.
5	**(d)** 0	She returns to the starting point.
6	**(a)** Newton's first law	Law of inertia.
7	**(a)** a force acts on a body and the body is displaced in the direction of the force	W = F x s.
8	**(a)** vibrating bodies	Vibration is the source.
9	**(b)** Robert Hooke	In thin slices of cork, 1665.
10	**(c)** tissue	Tissue.
11	**(b)** biodiversity	Diversity of life.
12	**(d)** asexual reproduction	No gametes involved.
13	**(c)** biosphere	Sphere of life.
14	**(c)** an approximate answer is enough for a decision	Orders of magnitude.
15	**(a)** a heterogeneous mixture whose particles settle on standing	Large particles.
16	**(c)** 4	Four atoms.
17	**(d)** protons	Z = p.
18	**(d)** acceleration	Change in velocity per time.
19	**(b)** are equal in magnitude, opposite in direction and act on different bodies	Third law.
20	**(c)** kinetic energy	W = change in KE.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Simplified representation of a real system (1); Ball-and-stick model of a molecule (1).	1+1
22	Homogeneous: uniform, e.g. salt solution (1); Heterogeneous: non-uniform, e.g. sand and water (1).	1+1
23	Mass is neither created nor destroyed in a chemical reaction (1); Antoine Lavoisier (1).	1+1
24	It could not explain the scattering of alpha particles (1); It had no nucleus (1).	1+1
25	Distance: total path length, scalar (1); Displacement: shortest straight-line gap with direction, vector (1).	1+1
26	A body remains at rest or in uniform motion unless acted on by an external force (1); Inertia is the tendency of a body to resist a change in its state (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Choose features important to the question (1); Ignore minor details such as air resistance for a falling ball (1); Assumptions make the problem manageable (1).	1+1+1
28	Heat the mixture: ammonium chloride sublimes (1); Add water to the residue, filter sand (1); Evaporate the filtrate to get salt (1).	1+1+1
29	Mass of reactants = 22.0 g (1); Mass of CO2 formed = 22.0 g (1); Mass before equals mass after (1).	1+1+1
30	Alpha particles directed at thin gold foil (1); Most passed; some deflected; very few rebounded (1); Nucleus small, dense and positive; mostly empty atom (1).	1+1+1

31	Distance = 3 + 4 = 7 m (1); Displacement = sqrt(3 squared + 4 squared) (1); = 5 m (1).	1+1+1
	OR: Net force = 5 N east (1); a = 5/2 (1); = 2.5 m/s squared (1).	1+1+1
32	Lifting: force and displacement same direction, positive (1); Lowering: force upward, displacement downward, negative (1); W = mgh = 60 J (1).	1+1+1
	OR: v = f lambda = 340 m/s (1); T = 1/200 (1); = 0.005 s (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Question from observation (1); Hypothesis and prediction (1); Fair test experiment (1); Data collection and analysis (1); Conclusion and further questions (1).	1+1+1+1+1
34	Pure: fixed composition (1); Mixture: variable (1); Gold: pure; air, sugar solution, brass: homogeneous (1); Milk: colloid, treated as heterogeneous (1); Properties differ (1).	1+1+1+1+1
35	Take both solutions in a flask with a small tube (1); Weigh the set-up (1); Mix: white precipitate of BaSO4 forms (1); Weigh again: mass is the same (1); Conclusion: mass is conserved (1).	1+1+1+1+1
36	Thomson: plum pudding (1); Drawback: could not explain alpha scattering (1); Rutherford: nucleus with revolving electrons (1); Drawback: instability (1); Bohr: fixed shells (1).	1+1+1+1+1
	OR: v = u + at from slope (1); s = area of trapezium = ut + (1/2)at squared (1); v squared = u squared + 2as by eliminating t (1); Meaning of symbols (1); Conditions: uniform acceleration (1).	1+1+1+1+1
37	First law with example of a coin on a card (1); Second law F = ma with a pushed trolley (1); Third law with rocket (1); Statements correct (1); Daily-life relevance (1).	1+1+1+1+1
	OR: W = F s cos(theta) (1); Along: positive (1); Opposite: negative (1); Perpendicular: zero (1); Example 60 J (1).	1+1+1+1+1
38	Vibrating source (1); Compression and rarefaction (1); Amplitude and wavelength (1); Frequency and time period f = 1/T (1); Wave equation v = f lambda (1).	1+1+1+1+1
	OR: Cell membrane (1); Nucleus with chromosomes (1); Cytoplasm with ER and Golgi (1); Mitochondria, lysosomes (1); Ribosomes and vacuoles (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** amplitude	Larger amplitude, louder.
2	**(d)** prokaryotic cell	e.g. bacteria.
3	**(d)** sclerenchyma	Dead, with lignin.
4	**(a)** algae	Thallophyta.
5	**(d)** stamen	Anther and filament.
6	**(a)** fresh snow	High albedo.
7	**(d)** a law describes what happens while a theory explains why	Different roles.
8	**(a)** pencil does not dissolve in the solvent	Ink would move.
9	**(b)** negatively charged ion	Gains electrons.
10	**(c)** 1	Gains 1 to make 8.
11	**(c)** velocity, acceleration and displacement	Third equation.
12	**(b)** one-sixth	g_moon is g/6.
13	**(a)** makes work easier by changing the magnitude or direction of force	Machines help us.
14	**(c)** 17.2 m	v t/2 = 344 x 0.1/2.
15	**(b)** rough ER	Protein synthesis.
16	**(c)** epithelial tissue	Covering.
17	**(d)** Porifera	Pores on the body.
18	**(b)** fruit	Fruit.
19	**(c)** stratosphere	About 15 to 35 km.
20	**(d)** a model	A visual model of matter.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Product of force and displacement in the direction of force (1); Joule (1).	1+1
22	By vibrating objects (1); A material medium (solid, liquid or gas) (1).	1+1
23	Robert Hooke in 1665 (1); Box-like compartments in a cork slice (1).	1+1
24	Group of similar cells doing a specific function (1); Division of labour makes bodies efficient (1).	1+1
25	Variety of living organisms on the Earth (1); Classification makes the study of the large variety easy (1).	1+1
26	Production of new individuals of the same species (1); Ensures the continuity of species (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Raisin in water swells due to endosmosis (1); Raisin in strong sugar solution shrinks due to exosmosis (1); Water moves through the selectively permeable membrane (1).	1+1+1
28	Grow onion roots in water (1); Measure the length daily (1); Growth takes place near the tip due to apical meristem (1).	1+1+1
29	Cell type (prokaryotic/eukaryotic) (1); Body organisation (unicellular/multicellular) (1); Mode of nutrition and presence of cell wall (1).	1+1+1
30	Nucleus divides into two (1); Cytoplasm divides (1); Two daughter Amoebae form (1).	1+1+1
31	Evaporation from water bodies (hydrosphere) (1); Condensation and rain from the atmosphere (1); Rain affects soil (geosphere) and plants (biosphere) (1).	1+1+1

	OR: Units were confused (litres and gallons or pounds and kilograms) (1); Wrong quantity of fuel was loaded (1); Care with units is vital (1).	1+1+1
32	12/100 x 250 (1); = 30 g (1); Water = 220 g (1).	1+1+1
	OR: Hard, brittle crystalline solids (1); High melting and boiling points (1); Conduct electricity when molten or dissolved (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Apical: tips, growth in length (1); Lateral: girth (1); Intercalary: base of leaves/internodes (1); Cells dividing, thin walls (1); Differentiate to form permanent tissues (1).	1+1+1+1+1
34	Monera (1); Protista (1); Fungi (1); Plantae (1); Animalia (1).	1+1+1+1+1
35	Fission (1); Budding (1); Fragmentation (1); Spore formation (1); Vegetative propagation (1).	1+1+1+1+1
36	Four spheres (1); Energy from the Sun (1); Matter cycles (1); Interactions (1); Human influence (1).	1+1+1+1+1
	OR: Instrument limits (1); Human errors like parallax (1); Repeat measurements (1); Take the average (1); Use better instruments (1).	1+1+1+1+1
37	Spot above the baseline (1); Solvent rises by capillary action (1); Components move at different rates (1); Compare heights (1); Used for dyes and pigments (1).	1+1+1+1+1
	OR: Oxygen = 8/3 x 18 = 48 g (1); Compound mass = 66 g (1); Compound is carbon dioxide (1); Ratio fixed by law of constant proportions (1); Conservation of mass verified (1).	1+1+1+1+1
38	K: p 19, n 20, e 19 (1); S: p 16, n 16, e 16 (1); F: p 9, n 10, e 9 (1); Atomic number identifies the element (1); Mass number gives nucleons (1).	1+1+1+1+1
	OR: Velocity along the tangent (1); Direction changes: accelerated (1); v = 2 x 3.14 x 0.5/2 (1); = 1.57 m/s (1); Speed constant (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** distilled water	Single compound.
2	**(b)** Joseph Proust	Proust.
3	**(b)** negative	Negative charge.
4	**(c)** greater than or equal to the magnitude of displacement	Path length is at least the straight-line gap.
5	**(d)** mass	More mass, more inertia.
6	**(d)** joule	1 J = 1 N m.
7	**(c)** a vacuum	A material medium is required.
8	**(d)** Schleiden and Schwann	Plants and animals are made of cells.
9	**(b)** striated (voluntary) muscles	Skeletal.
10	**(a)** gills	Aquatic.
11	**(a)** testes	Male organ.
12	**(b)** water (hydrological) cycle	Cycle of water.
13	**(c)** 10000 litres	0.5 x 20000.
14	**(d)** fog	Water droplets in air.
15	**(b)** 8	Eight atoms.
16	**(b)** protons plus neutrons	A = p + n.
17	**(c)** displacement	Product of v and t.
18	**(c)** act on different bodies	Different bodies.
19	**(a)** can neither be created nor destroyed but only changed from one form to another	Law.
20	**(c)** frequency	Higher frequency, higher pitch.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Geosphere: rocks and soil; hydrosphere: water (1); Atmosphere: gases; biosphere: life (1).	1+1
22	It is a deliberate assumption to simplify (1); The model is accurate enough for the purpose (1).	1+1
23	Separation by spinning at a high speed (1); Cream from milk, blood cells from plasma (1).	1+1
24	Number of atoms in a molecule (1); 2, 3 and 4 (1).	1+1
25	Protons = electrons = 11 (1); Neutrons = 12 (1).	1+1
26	Rate of change of velocity (1); m/s squared (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Isotopes: same Z, different A (1); Isobars: same A, different Z (1); Isotopes have the same chemical properties (1).	1+1+1
28	Graph is a horizontal line (1); Area = 6 x 10 (1); = 60 m (1).	1+1+1
29	a = 3 m/s squared (1); v = at = 9 m/s (1); Starts from rest (1).	1+1+1
30	Energy = 0.1 kW x 10 h = 1 kWh (1); 1 unit (1); Cost = 6 rupees (1).	1+1+1
31	Pulse sent and its echo is received (1); Depth = v t/2 (1); = 1500 x 4/2 = 3000 m (1).	1+1+1
	OR: Shape suits the function (1); RBC: disc-shaped, carries oxygen (1); Nerve cell: long and branched, carries messages (1).	1+1+1
32	Cell body with nucleus (1); Dendrites receive impulses (1); Axon carries the impulse away (1).	1+1+1

	OR: Porifera: pores, canal system, sessile (1); Cnidaria: stinging cells, radial symmetry (1); Examples: Sycon; Hydra (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Net force 20 N (1); a = 4 m/s squared (1); v = 16 m/s (1); s = 32 m (1); Formulas used (1).	1+1+1+1+1
34	At the extreme: PE max, KE zero (1); At the mean: KE max, PE zero (1); Total constant (1); Air resistance reduces amplitude (1); Energy converted to heat (1).	1+1+1+1+1
35	Repeated reflections (1); Persists and blurs speech (1); Absorbing materials: curtains, carpets (1); Curved sound boards reflect sound to the audience (1); Roof design (1).	1+1+1+1+1
36	Mitosis for growth and repair (1); Meiosis for gamete formation (1); Chromosome numbers (1); Daughter cells (1); Significance (1).	1+1+1+1+1
	OR: Striated (1); Smooth (1); Cardiac (1); Voluntary or involuntary (1); Locations (1).	1+1+1+1+1
37	Cotyledons (1); Roots (1); Leaf venation (1); Flower parts (1); Examples (1).	1+1+1+1+1
	OR: Parents (1); Gametes (1); Variation (1); Speed (1); Examples (1).	1+1+1+1+1
38	Fixation by bacteria and lightning (1); Absorption of nitrates (1); Passing through the food chain (1); Decay by decomposers (1); Return to the air (1).	1+1+1+1+1
	OR: Ball and stick model (1); Atomic models (Dalton to Bohr) (1); Ideal gas or frictionless motion (1); New evidence leads to revision (1); Models are useful, not final (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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